



APEX ENGINEERING GROUP
Engineering Flow, Designing Solutions

CAD Designer

Company Profile

We are APEX ENGINEERING GROUP, an engineering consultancy specializing in architectural design, civil engineering solutions, structural planning, and 3D modeling services.

We support clients in residential, commercial, and industrial projects by providing innovative, efficient, and practical design solutions tailored to project requirements. Our expertise includes AutoCAD drafting, 3D visualization, structural detailing, planning layouts, and construction oriented design support.

Attached are projects demonstrating our technical approach, design capabilities, and the quality standards we deliver to clients.

We would be pleased to support your project with professional, high-quality engineering and architectural solutions.

Best

APEX ENGINEERING GROUP

regards,

About Us

We have specialization in CFD analysis, thermal simulations, and engineering design. We work on industrial flow and heat transfer problems using simulation tools to improve product performance, airflow behavior, and thermal efficiency. Our expertise includes ANSYS Fluent, CAD modeling, engineering analysis, and technical reporting.

Services

Design & Modeling

- 3D Mechanical Modeling
- Product Design
- Industrial Component Design
- Assembly Modeling
- Auto Cad
- Solid Modeling

Engineering Support

- Technical Reports
- Simulation Reports
- Research & FYP Support
- Design Optimization
- Industrial Problem Solving

Project 1

Quenching Chamber

Industry

Polymer Processing Industry

Problem Statement

The client was experiencing inconsistent cooling performance within a quenching chamber, leading to uneven product quality and reduced manufacturing efficiency. The existing chamber design lacked optimized internal geometry, resulting in non-uniform distribution of flow and temperature.

Objective

To improve the quenching chamber design by optimizing internal geometry and structural layout to achieve uniform cooling, improved efficiency, and consistent product quality.

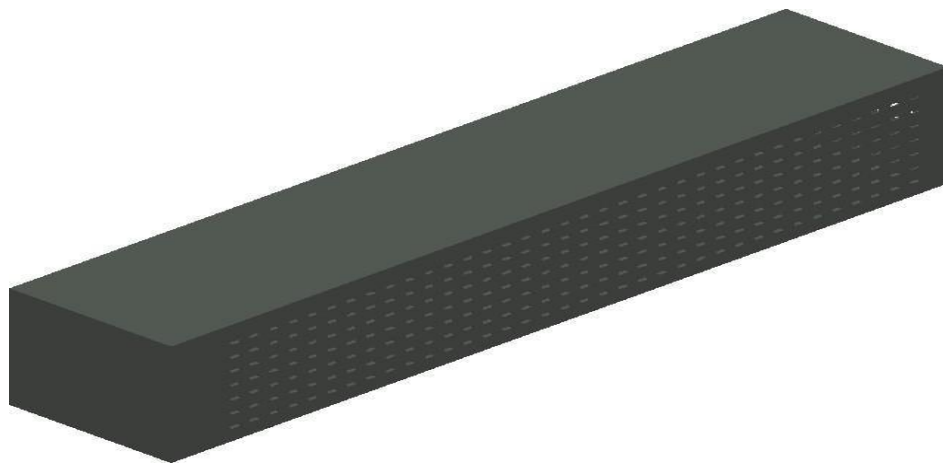


Figure 1 geometry

Software Used

- SolidWorks

Approach & Methodology

- 3D modeling of quenching chamber in SolidWorks
- Detailed geometric evaluation of internal structure
- Design review for improving internal flow paths and distribution zones
- Modification of chamber geometry for improved uniformity
- Iterative design refinement for better performance output

Design Findings

- Uneven distribution caused by internal geometric constraints
- Existence of low-performance zones affecting overall efficiency
- Non-uniform internal conditions due to design limitations
- Need for optimized structural configuration for balanced performance

Design Improvements & Impact

- Improved internal geometry for better distribution behavior
- Enhanced design uniformity through SolidWorks-based modeling
- Reduced inefficiencies by optimizing structural layout
- Improved consistency in product performance
- Strengthened overall engineering design quality

Conclusion

The original chamber design had limitations in achieving uniform internal performance due to geometric constraints. Through SolidWorks based 3D modeling and design refinement, improvements were implemented in the internal structure, resulting in a more balanced, efficient, and production friendly design.

Project 2

Industrial Design Project

Finned Tube Heat Exchanger

Industry

HVAC / Heat Transfer Systems

Project Overview

This project focuses on the 3D CAD design and modeling of a finned tube heat exchanger used in thermal management systems. The design was developed to study geometry, fin arrangement, and tube layout for improved heat transfer efficiency and manufacturability.

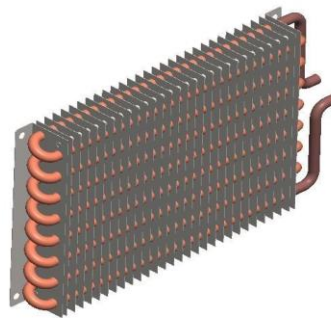


Figure 1: Geometry of finned tube heat exchanger

Objective

To develop an accurate 3D model of a finned tube heat exchanger using CAD tools and evaluate its design for effective thermal performance, structural alignment, and practical manufacturing feasibility.

Software Used

- SolidWorks

Approach & Methodology

- Development of detailed 3D model of finned tube assembly in SolidWorks
- Design of tube routing and fin geometry for optimized layout
- Assembly modeling with proper part alignment and constraints
- Structural arrangement for uniform spacing and compact design
- Focus on manufacturable and industry-standard design practices

Design Features

- Precisely arranged finned tube configuration
- Optimized spacing between fins for effective heat transfer
- Compact and efficient coil layout design
- Accurate tube bending and routing in 3D environment
- Fully assembled model with realistic engineering constraints

Design Insights

- Proper fin arrangement improves surface area utilization
- Compact coil structure enhances system efficiency
- Accurate geometric modeling ensures manufacturability
- Assembly constraints help maintain structural stability
- Design supports effective integration in HVAC systems

Design Impact

- Improved understanding of heat exchanger geometry design
- Enhanced CAD modeling and assembly skills using SolidWorks
- Provided manufacturable and industry-ready design output
- Reduced dependency on physical prototyping through digital modeling.
- Strengthened design optimization approach for thermal systems

Conclusion

The finned tube heat exchanger was successfully designed using SolidWorks, focusing on accurate geometry, efficient fin arrangement, and realistic assembly modeling. The final CAD model represents a manufacturable and optimized design suitable for HVAC and heat transfer applications.

Project 3

Aerodynamic Design Project

Formula 1 Front Wing

Industry

Automotive Aerodynamics / Motorsport Design

Project Overview

This project focuses on the 3D CAD design and aerodynamic configuration development of a Formula 1 front wing. The study includes multi-element wing geometry modeling with different flap angles and slot gap arrangements to evaluate aerodynamic behavior and design efficiency.

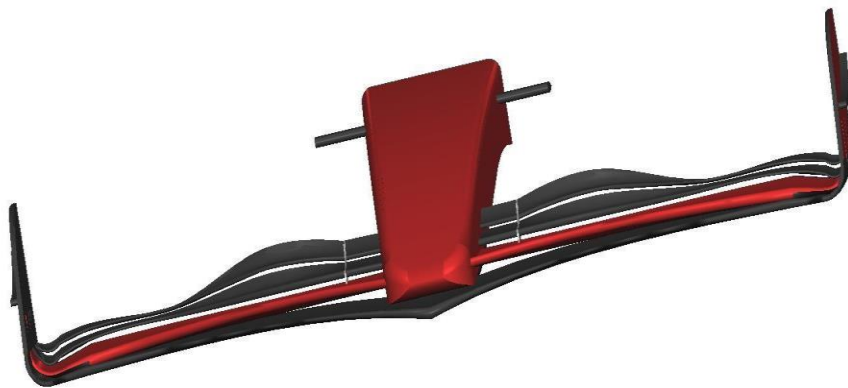


Figure 1: Front Wing Geometry

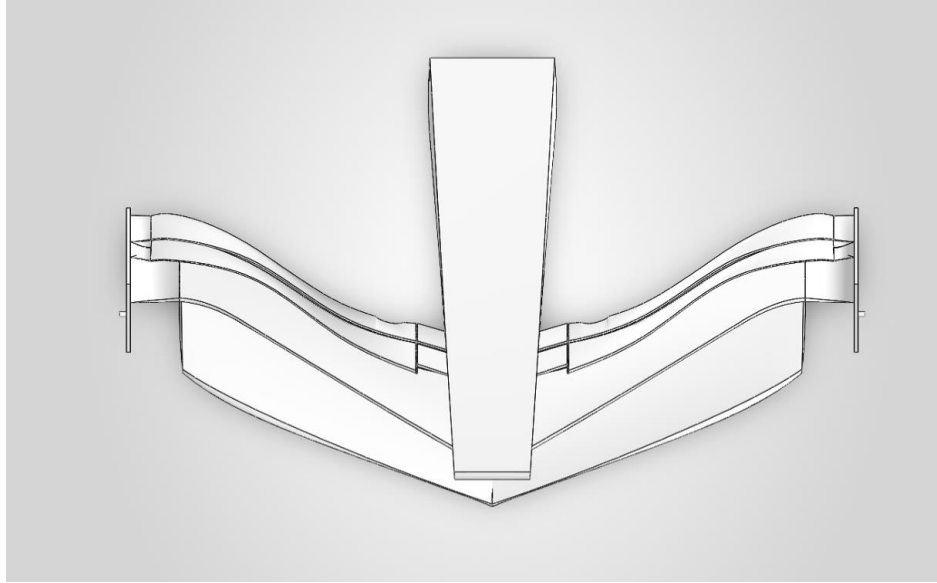


Figure 2: Top View Front Wing Geometry

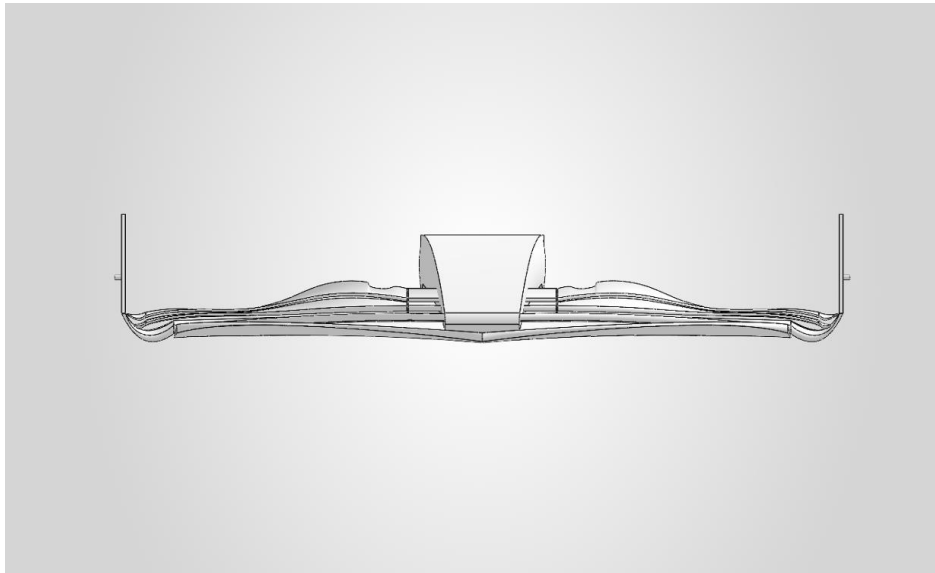


Figure 3: Front View of Front Wing Geometry

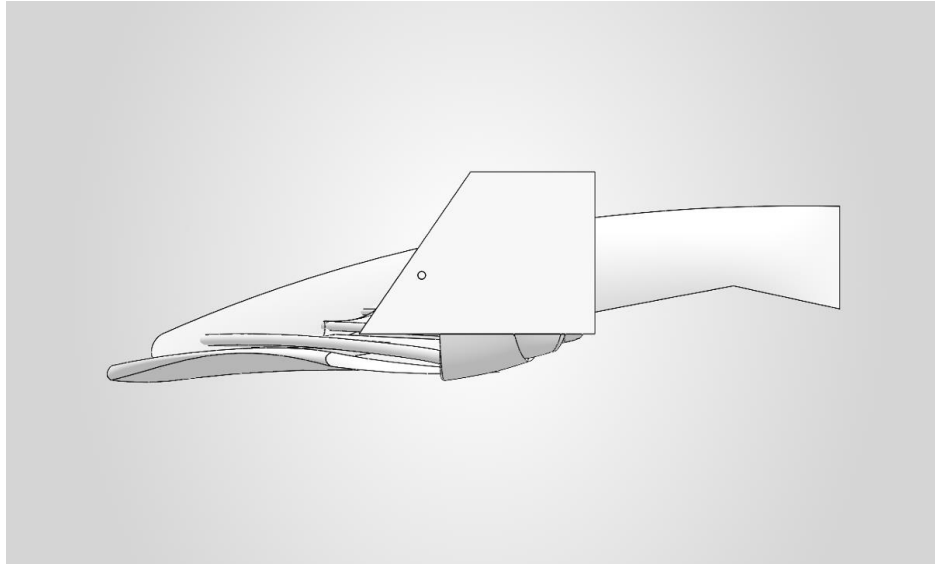


Figure 4: Side View of Front Wing Geometry

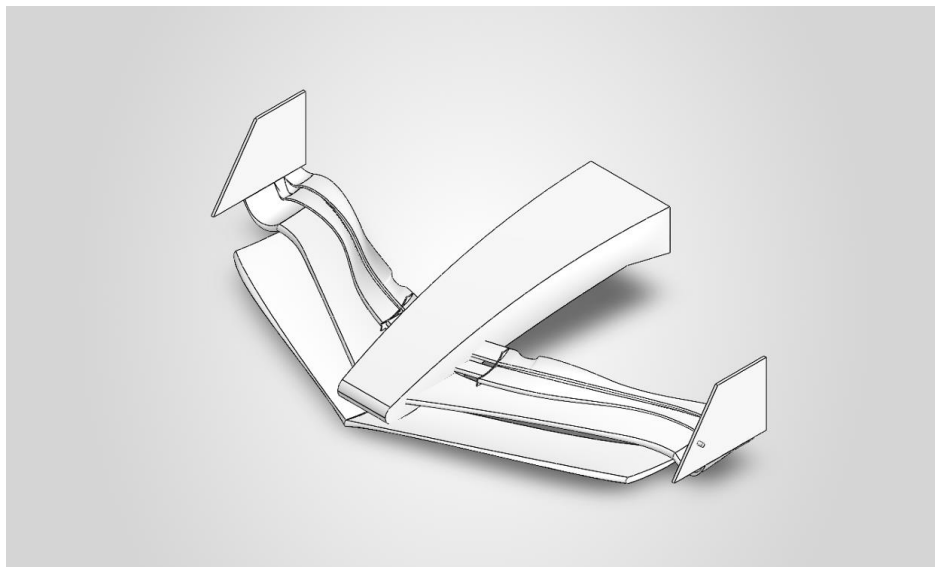


Figure 5: Isometric View of Front Wing Geometry

Objective

To design and optimize a multi element front wing using CAD tools by analyzing different geometric configurations of flap angles and slot gaps to improve aerodynamic performance and flow behavior.

Software Used

- SolidWorks

Approach & Methodology

- 3D modeling of multi-element front wing in SolidWorks
- Development of different flap angle configurations
- Variation in slot gap geometry for performance comparison
- Assembly modeling with accurate alignment of aerodynamic elements
- Design-based evaluation of airflow behavior and efficiency trends
- Comparative study of multiple CAD configurations

Design Findings

- Flap angle significantly influences airflow behavior across wing elements
- Slot gap variation affects interaction between aerodynamic surfaces
- Optimized geometry improves flow consistency and energy transfer
- Balanced configurations provide improved aerodynamic stability
- Design sensitivity observed with small geometric changes

Results & Design Impact

- Identified optimized flap angle and slot gap configuration through design iteration
- Improved understanding of multi-element aerodynamic design behavior
- Achieved enhanced aerodynamic efficiency through geometry optimization
- Best configuration showed improved lift performance with controlled flow behavior
- Supported design decisions before physical prototyping stage

Conclusion

The multi-element front wing was successfully designed using SolidWorks with multiple geometric configurations to evaluate aerodynamic performance trends. The study highlights how flap angle and slot gap variations influence overall design efficiency. The optimized CAD configuration provided improved aerodynamic behavior and serves as a strong foundation for advanced motorsport design development.

Project 4

CFD Study of Battery Spacing and Airflow for Efficient Cooling of Li-ion Packs

Project Overview

This Project focuses on the design and thermal optimization of lithium-ion battery pack architecture with different cell spacing configurations and airflow layouts. The study was carried out to improve cooling efficiency, temperature uniformity, and airflow distribution within the battery module using advanced CAD based modeling and engineering analysis.

The design includes evaluation of straight-type and Z-type airflow configurations along with 1 mm, 2 mm, and 3 mm cell spacing variations to achieve an optimized thermal management solution.

Objective

To develop and evaluate optimized battery pack designs by analyzing different cell spacing and airflow configurations in order to enhance cooling performance, reduce hotspot formation, and improve overall thermal stability.

Software Used

- SolidWorks

Design Methodology

- 3D modeling of lithium-ion battery pack geometry in SolidWorks
- Development of straight-type airflow configuration design
- Development of Z-type airflow configuration design
- Modeling of cell spacing variations: 1 mm, 2 mm, and 3 mm

- Assembly creation of full battery module with accurate alignment
 - Design-based thermal behavior evaluation through engineering simulation setup
- Comparative study of multiple design configurations

Design Configurations Studied

- Straight-Type airflow arrangement
- Z-Type airflow arrangement
- Cell spacing variations:
 - 1 mm spacing
 - 2 mm spacing
 - 3 mm spacing

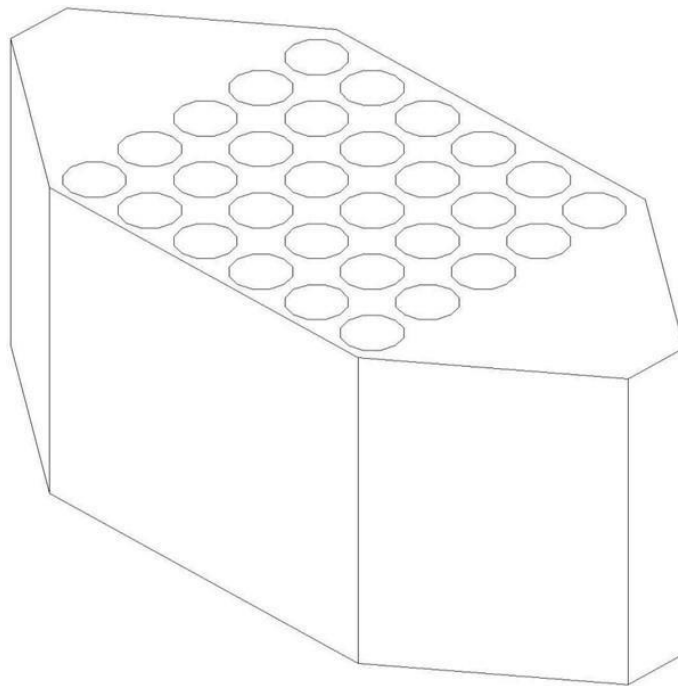


Figure 1: Isometric View of Lithium-ion battery pack of Straight Type

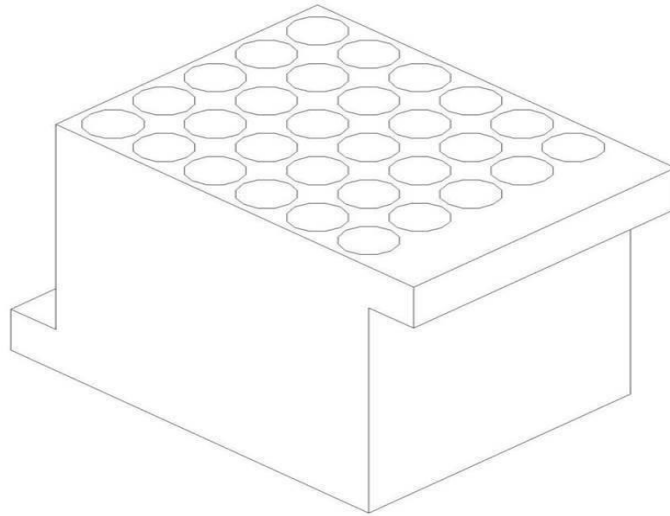


Figure 2: Isometric View of Z-Type Lithium-ion battery pack

Key Design Findings

- Cell spacing significantly affects airflow distribution inside the battery pack
- Z-type configuration improves airflow uniformity compared to straight-type design
- Smaller spacing (1 mm) increases thermal interaction but may restrict airflow
- Larger spacing (3 mm) improves airflow but reduces packing density efficiency . 2 mm spacing provides balanced thermal and structural performance

Engineering Results

- Improved airflow distribution with optimized spacing and geometry
- Reduced hotspot formation inside battery pack design
- Lower flow resistance at higher spacing configurations
- Better thermal uniformity achieved using Z-type airflow layout
- Design validation performed using engineering correlation methods (Reynolds & Nusselt based comparison).

Skills Demonstrated

- 3D CAD Design & Assembly Modeling
- Thermal System Design Understanding
- Engineering Simulation Setup
- Heat Transfer & Fluid Flow Interpretation
- Data Visualization & Technical Reporting
- Design Optimization for Real-World Applications

Conclusion

The project successfully demonstrates the importance of cell spacing and airflow configuration in lithium-ion battery pack design. Through SolidWorks-based modeling and comparative design analysis of straight-type and Z-type airflow systems with 1 mm, 2 mm, and 3 mm spacing, an optimized configuration was identified that improves thermal uniformity, reduces hotspots, and enhances overall battery performance.

Project 5

Aerodynamic Aircraft Wing & Rib Structure Design

Project Overview

This project showcases the advanced 3D CAD modeling of an Aircraft Wing Component featuring internal structural reinforcement designed in SolidWorks. The model focuses on capturing precise aerodynamic profiles (Airfoils) and translating them into a lightweight, high-strength structural design ready for manufacturing or Computational Fluid Dynamics (CFD) grid generation.

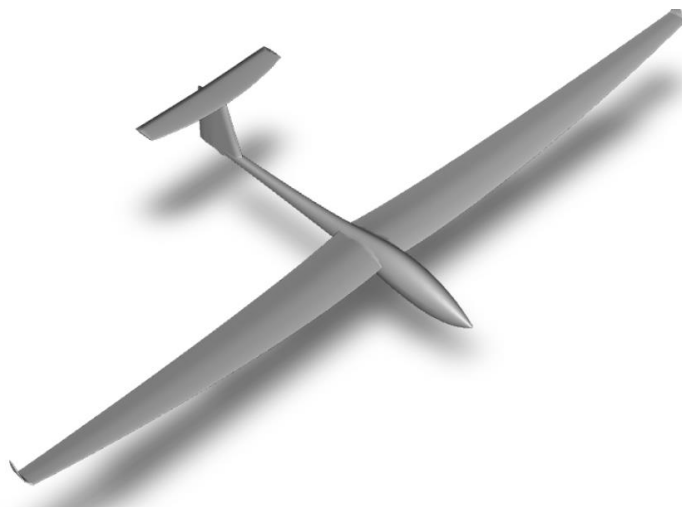


Figure 1: SolidWorks 3D CAD model of the aircraft wing and airfoil profile

Technical Specifications & CAD Features

- **Software Used:** SolidWorks (Advanced Part & Surface Modeling)
- **Design Features:** Smooth aerodynamic skin profile based on standard airfoil coordinates.
 - Internal hollow pocketing/rib structures to optimize strength to weight ratio (Aerospace lightweighting techniques).
 - Integrated mounting tabs and clearance slots for structural fastening.
- **CAD Methodologies:** Extruded cuts with precise draft angles, thin-feature modeling, and parametric sketch constraints.

Engineering Skills & Core Competencies Demonstrated

1. Aerodynamic Geometric Modeling

- Successfully lofted/extruded complex curved surfaces while maintaining profile accuracy, which is essential for minimizing drag and optimizing lift characteristics in aerospace applications.

2. Aerospace Structural Design (Lightweighting)

- Demonstrated the application of internal webbing and structural pockets. This mimics real-world aviation design where material is removed from low-stress regions to reduce the overall weight of the aircraft without compromising structural integrity.

3. Manufacturing & Simulation Readiness

- Built with clean geometric constraints making the model perfectly optimized for downstream engineering workflows, such as **ANSYS Fluent (CFD Analysis)** for lift/drag evaluation, or structural FEA testing.

“Thank you for your time”

For orders, or project inquiries, please feel free to contact:

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